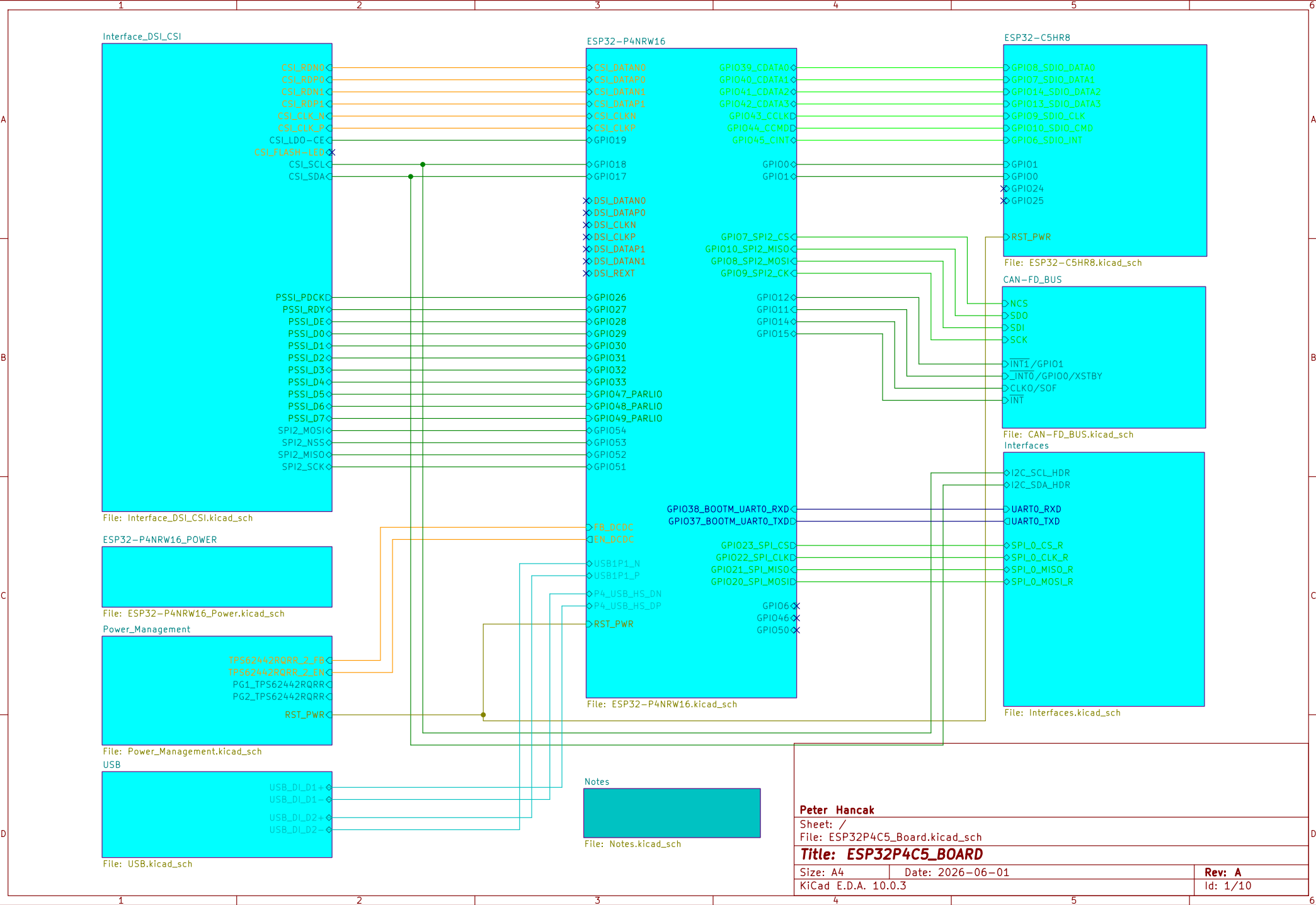
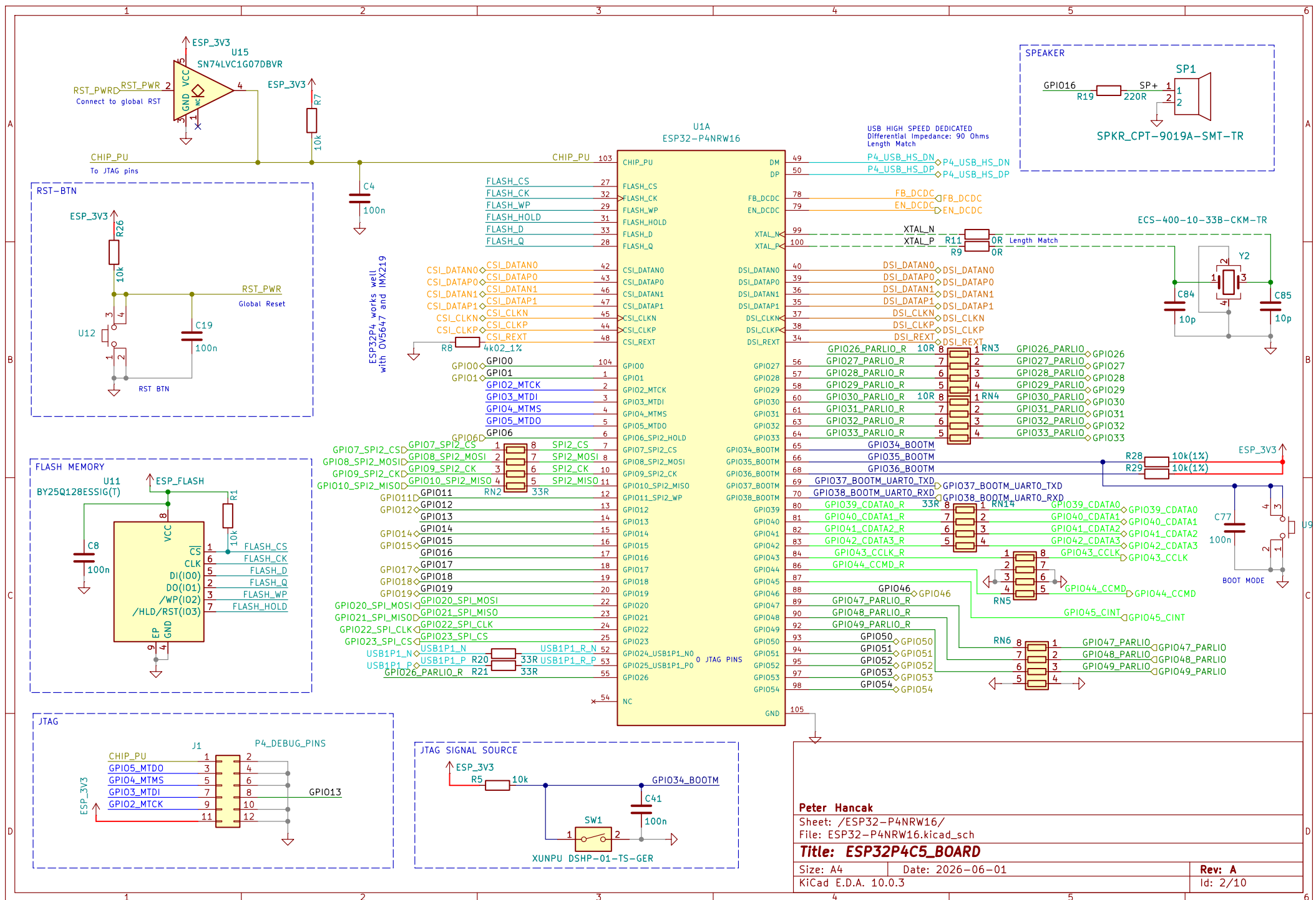
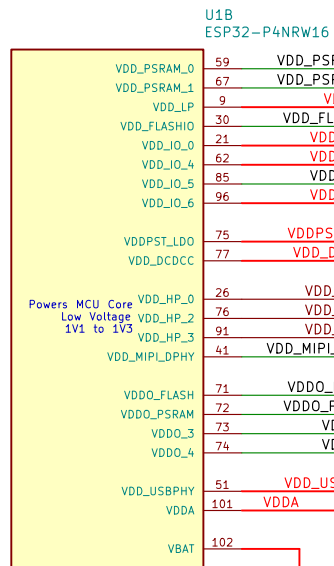
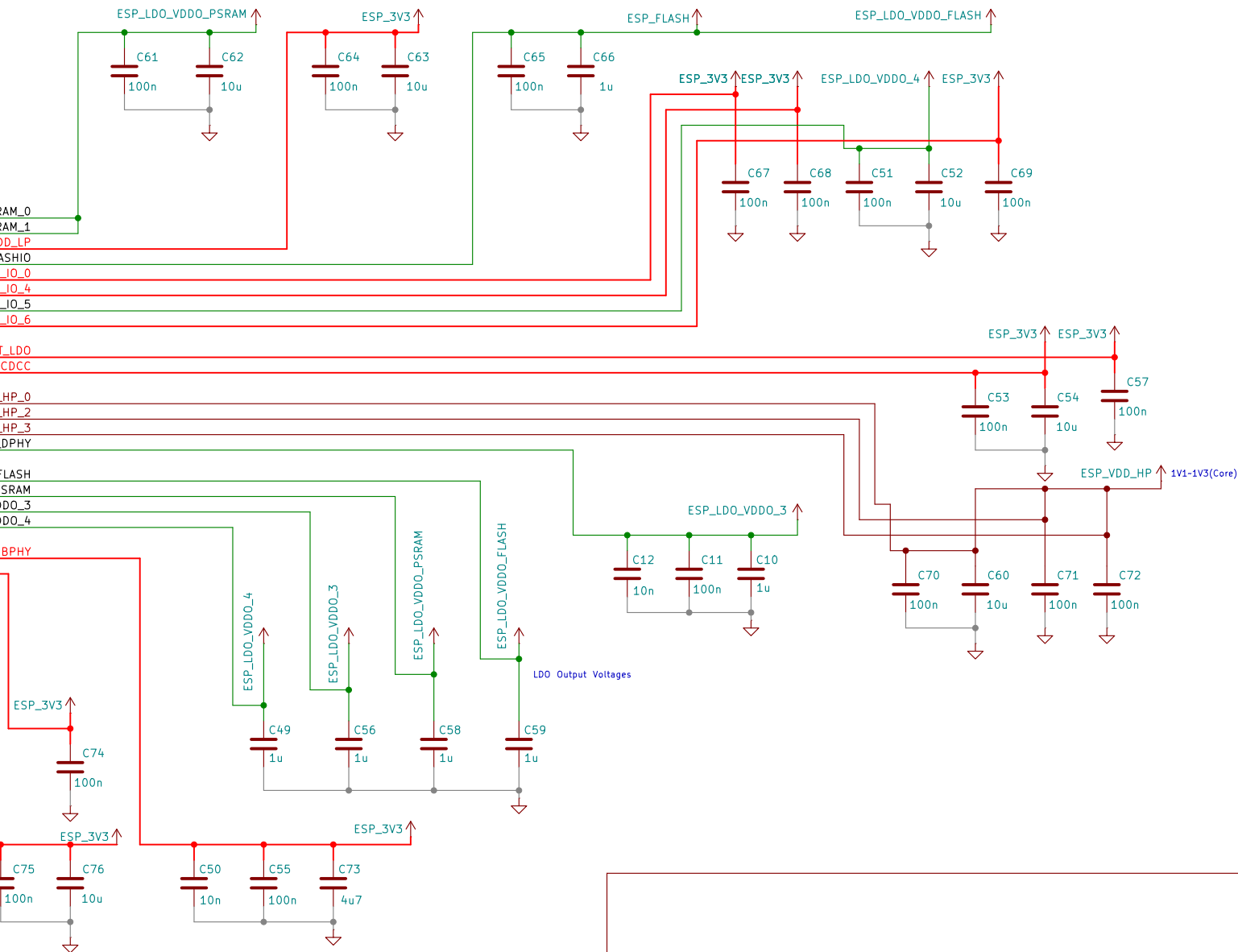








ESP_3V3 is from 5V LDO or SMPS



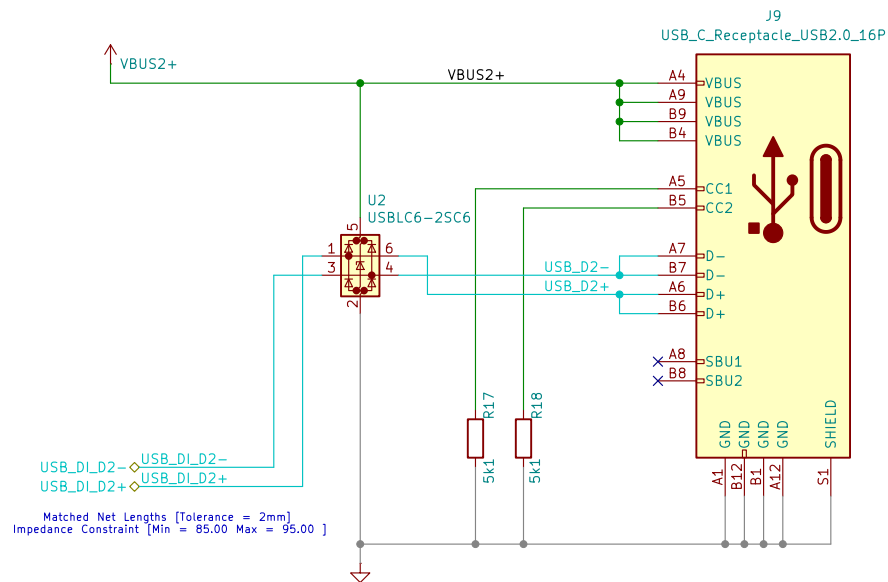
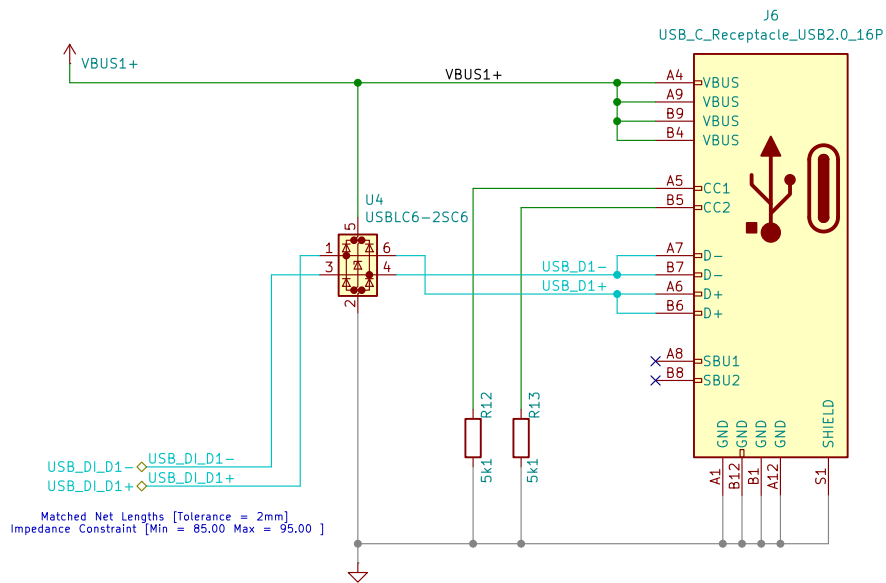
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Sheet: /ESP32-P4NRW16_POWER/
File: ESP32-P4NRW16_Power.kicad_sch

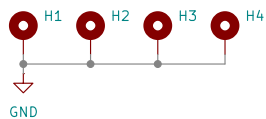
Title: **ESP32P4C5_BOARD**

Size: A4 Date: 2026-06-01
KiCad E.D.A. 10.0.3

Rev: A
Id: 3/10



Mounting Holes



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Sheet: /USB/

File: USB.kicad_sch

Title: ESP32P4C5_BOARD

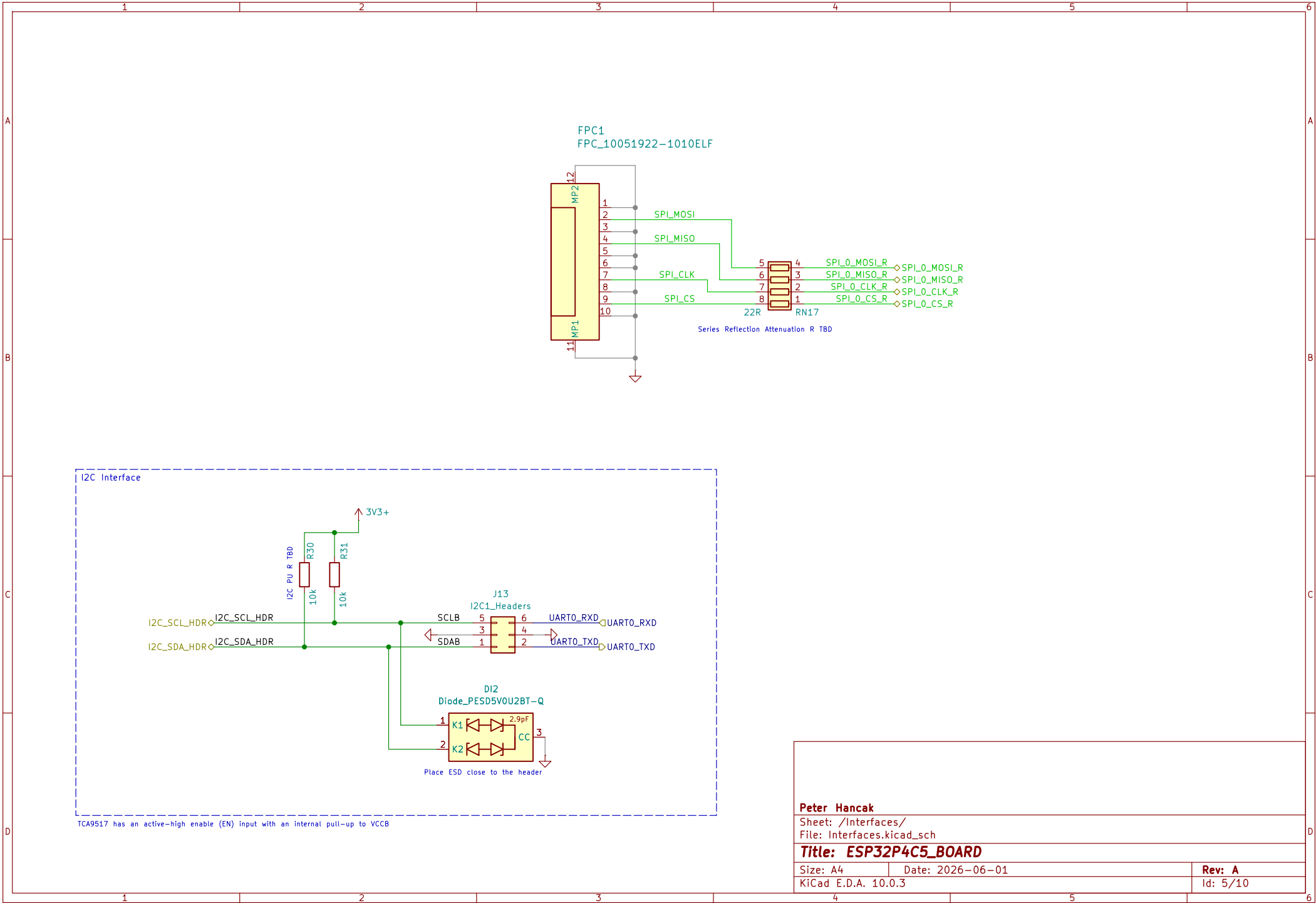
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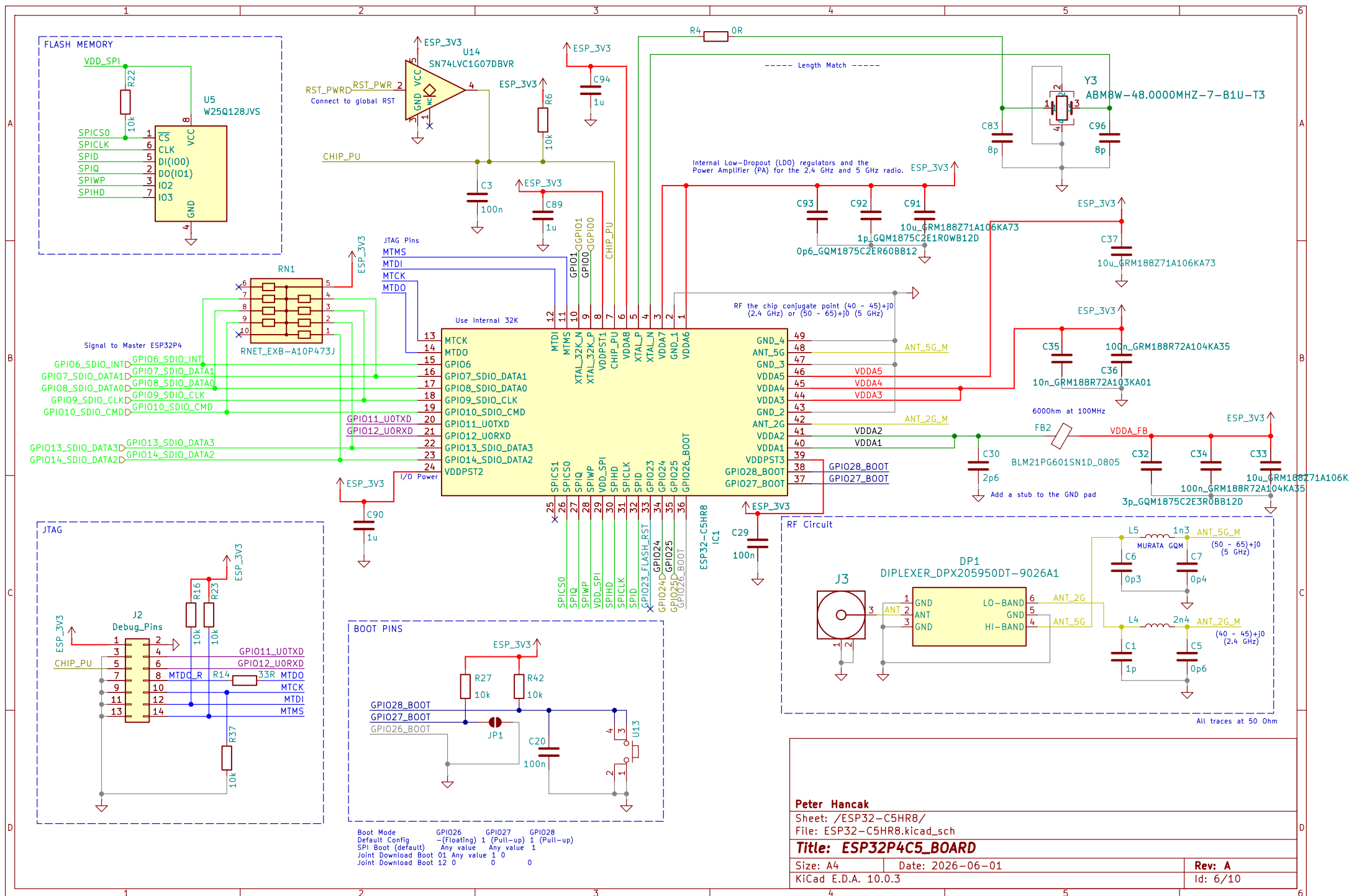
Date: 2026-06-01

Rev: A

KiCad E.D.A. 10.0.3

Id: 4/10





Peter Hancak

Sheet: /ESP32-C5HR8/

File: ESP32-C5HR8.kicad_sch

Title: ESP32P4C5_BOARD

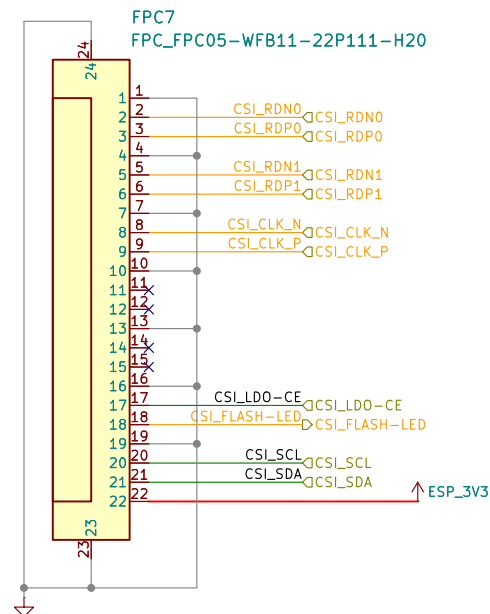
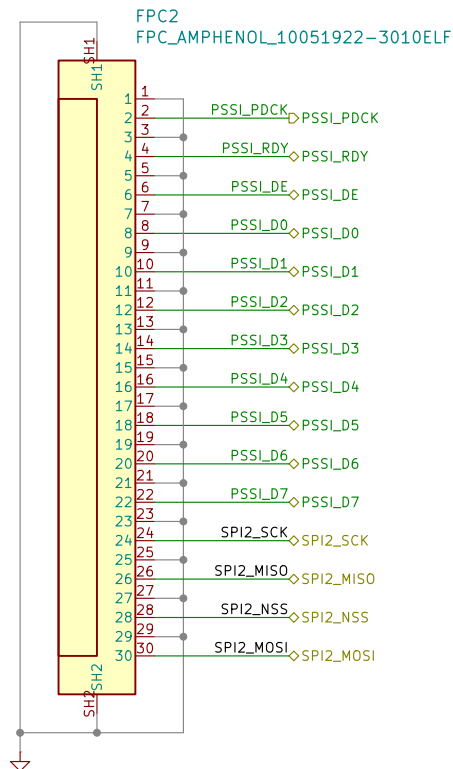
Size: A4

Date: 2026-06-01

KiCad E.D.A. 10.0.3

Rev: A

Id: 6/10



FPC REQUIREMENTS/GOALS:
- based on 54548-227
0.50mm Pitch Easy-On FFC/FPC Connector, 1.20mm Height,
Right-Angle, Surface Mount, ZIF, Bottom Contact Style, 22 Circuits,
Gold (Au)Plating0.50mm Pitch Easy-On FFC/FPC Connector,
1.20mm Height, Right-Angle Surface Mount, ZIF, Bottom Contact
Style, 22 Circuits, Gold (Au)Plating

MIPI 2 LANE 40PIN 0.5P REFERENCE
https://www.alibaba.com/product-detail/3-5-Inch-LCD-480-800_1601636999524.html?spm=a2700.proseach.normal_offer.d_image.311067af89LjnB&priceId=2b7e3e7b4ca24f6aad4eb9f0880c6bb

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Sheet: /Interface_DSL_CSI/
File: Interface_DSL_CSI.kicad_sch

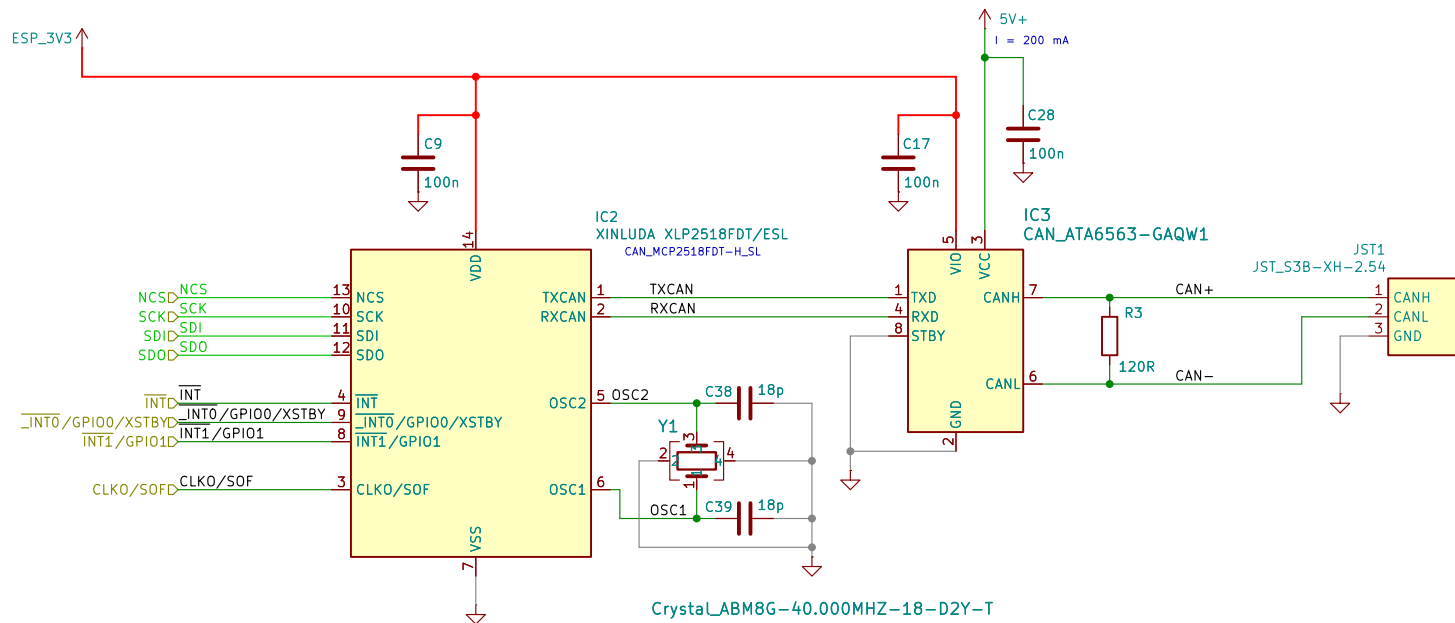
Title: **ESP32P4C5_BOARD**

Size: A4 Date: 2026-06-01

KiCad E.D.A. 10.0.3

Rev: A

Id: 8/10



Use INT0 when your software needs to know: "Is there data for me to read and process?"
 Use SOF when your software needs to know: "Exactly what microsecond did this network event happen in physical reality?"

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Sheet: /CAN-FD_BUS/
 File: CAN-FD_BUS.kicad_sch

Title: ESP32P4C5_BOARD

Size: A4 Date: 2026-06-01
 KiCad E.D.A. 10.0.3

Rev: A
 Id: 9/10

	1	2	3	4	5	6			
A	POWER MANAGEMENT NOTES: The input power goes through three stages 1. a) External Power: LM73100RPWR, 2.7 – 23 V, 5.5 A Integrated Ideal Diode with Input Reverse Polarity and Overvoltage Protection b) USB POWER: SS34–E3/57T Diode 2. TPS259531DSGR 2.7V to 18V, 4–A, 34–mQ eFuse With Fast Overvoltage Protection 3. TPS62442RQRR, 2.75–V to 6–V, Dual–Channel Step–Down Converter with Adjustable Frequency – Stage Description: 1. The power is sourced from the immediately connected power source without relying on jumpers to manual selection. The diodes prevent current from flowing back to a source in case of multiple power sources being connected. The ideal diode was used for the external power source to limit power loss inherent in a regular diode as the eternal power source will at times be a battery. The ideal diode includes polarity protection (not required on the USB sources) 2. The eFuse provides overvoltage protection, undervoltage lockout, Short Circuit Protection, and limits inrush current. 3. SMPS that outputs two power rails. The first is dedicated and controlled by the ESP32P4 core. The second is general 3.3V power rail used by all other components. ESP32P4 Power Domains – THE VDD_HP pins: 600mA to 1A Current, Required Width: 0.3 mm (1A) Ideal Diode LM73100RPWR Parameter Notes: – Maximum System Peak Current (IOUT): 2.5 A (to cleanly capture joint ESP32 radio bursts, camera operations, and converter losses). – I IMON is not used (connected to GND), no monitoring pins available on the MCU – Calculate the Value for RIMON = $VADC_MAX/IIMON = 3.3\text{ V}/452.5\text{ }\mu\text{A} \approx 7.29\text{ k}\Omega$ IMON EFUSE TPS259531DSGR Notes: – Short circuit protection/current limiter – Max current: R_ILM = 387 Ohms (500 Ohm) at 4.17 A – Over Voltage Clamp (OVC) set at 5.7V (for TPS25953x) SMPS TPS62442RQRR Notes: – The PG1 and PG2 left floating as they are not used. The MCU PU pins are handled through a parallel C.			CONTROLLED IMPEDANCE TRACE WIDTHS (Simulations done using ADS Controlled Impedance Line Designer) – Simulated is JLCPCB: JLC04161H–3313(Finished thickness1.56mm±10%) 1. Microstrip Edge Coupled, 100 Ohm Differential Impedance, 0.35mm (center to center), 50mmm Length, f=1Ghz, Width = 0.1225 mm 2. Coplanar Edge Coupled, 100 Ohm Differential Impedance, 0.35mm (center to center), 0.25mm Clearance, 50mmm Length, f=1Ghz, Width = 0.1354 mm, 0.35 – 0.1354 = 0.2146 mm (edge to edge spacing) 3. USB Coplanar Edge Coupled, 90 Ohm Differential Impedance, 0.5mm (center to center), 0.25mm Clearance, 50mmm Length, f=240Mhz, Width = 0.1704 mm, 0.5 – 0.1704 = 0.3296 mm 4. USB Microstrip Edge Coupled, 90 Ohm Differential Impedance, 0.5mm (center to center), 50mmm Length, f=240Mhz, Width = 0.1722 mm RF TRACE 50 Ohm – Coplanar single–ended , f is 2.5 GHz, L 20mm, Clearance 0.25mm, Width = 0.1472 mm – Coplanar single–ended , f is 5 GHz, L 20mm, Clearance 0.25mm, Width = 0.1475 mm SMA CONNECTOR SIGNAL PIN (MIDDLE) FOOTPRINT – With normal 0.35 mm clearance resulted in a very significant signal degradation – I recalculated the clearance for the middle pin using ADS: – Coplanar Single–Ended, Length: 5mm, Width: 1.37 mmm (width of the pin), f: 5GHz, Clearance = 0.4046 mm ESP32C5 DESIGN CONSIDERATIONS: – Flash memory traces NOT length matched. (at 80 MHz not needed) – Isolate the CLK trace on the ext flash memory bus CAN–FD INTERFACE: – Traces CANH, CANL impedance matched to 120 Ohm – 120 Ohm shunted across CANH and CANL (to be done on the receiver and sender) SERIES TERMINATION – CSI, DSI, SDIO (near P4), UART0, UART1, NOT I2C SDIO – Array Series termination on SDIO DATA 0–3 – Individual series termination on CLK, CMD – PU on the SDIO INT (Interrupt) pin (Open Drain)			ESP32C5 SETUP NOTES: 1. Set GPIO0 through GPIO MATRIX to GPIO (From 32K XTAL) 2. Set GPIO1 through GPIO MATRIX to GPIO1 (From 32K XTAL) 3. To enable programming through the JTAG pins eFuse 2: EFUSE_DIS_USB_JTAG has to be set to 1 (default 0) – Set BOOT pins to Joint Download Boot 1 mode (to download through JTAG pins) – In Joint Download Boot 1 mode, the USB Serial/JTAG controller is forcibly disabled, and the JTAG signal only comes from JTAG pins.		
B	PROJECT REQUIREMENTS/GOALS: – Camera OV5647 to capture real–time vide (or similar) – Provide WIFI link (to transmit the real–time video) – Control motor through CAN–FD interface – High Speed 8–bit pparallel connection to expansion PCBs – High Speed USB connector – I2C, UART, SPI (through FPC) interfaces SECONDARY GOALS: – Function as a simple server – Motor control through PWM pins – LCD with touch screen (MIPI 2LANE)								
C	ESP32C5 HARDWARE CONSIDERATIONS: – The C5 does not have a USB connector – Programming is to be done through JTAG pins – To enable programming through the JTAG pins eFuse 2: EFUSE_DIS_USB_JTAG has to be set to 1 (default 0) – BOOT CONFIGURATION – GPIO26_BOOT is pulled to GND (for all boot configurations) – GPIO27_BOOT and GPIO28_BOOT have PU R with a pull–down BTN – Solder bridge allows for either both to be pulled LOW or only the GPIO28_BOOT						ESP32P4 SETUP NOTES: 1. The PARLIO (Parallel Interface) pins are set as general PARLIO IO to be set up with specific pin functtlin through the GPIO Matrix. The function of each PARLIO pin can be determined based on the target device. 2. GPIO45 is to be set up as CINT (SDIO Interrupt) for the ESP32C5 to let the P4 know of available information on the SDIO bus. 3. The W25Q256JWPIQ flash memory is a 1.8V device and is powered as well as logic I/O from VDD0_FLASH (PIN71). Set up logic 1.8V output before connecting. Set the EFUSE_0PXA_TIEH_SEL0 register. (DO NOT solder on the flash until the flash votage is set to 1.8V) 4. VDD_HP_n is CORE power with 1.3V MAX. Set to 1.2V (as per reference) 5. BOOT CONFIGURATION – The ESP32P4 can be programmed using USB–Serial–JTAG Download (The USB Serial/JTAG Controller is able to put the ESP32–P4 into download mode automatically – NO NEED TO PULL GPIO35 LOW MANUALLY) – GPIO35 HIGH: SPI Boot mode (default). PU is permanently on GPIO35 – GPIO35 LOW: Joint Download Boot mode (USB–Serial–JTAG Download Boot, UART Download Boot, SPI Slave Download Boot, USB 2.0 OTG Download) – JTAG Signal Source Control – eFuse 3: EFUSE_JTAG_SEL_ENABLE to 1 – GPIO34 can be used for JTAG signal source – 1: JTAG signals come from USB Serial/JTAG Controller – 0: JTAG signals come from corresponding pins MTDI, MTCK, MTMS, and MTDO		
D	ESP32P4 HARDWARE CONSIDERATIONS: – DEBUG – The P4 has both USB ports exposed and the UART0 as a last resort debug. – Main flash/debug is to be done through the USB/JTAG port – The MCU JTAG pins are exposed on a header but do not include the PU and PD resistors – BOOT CONFIGURATION – CS_LDO–CE Pin on the CAMERA turns the internal camera LDO ON or OFF thereby allowing the chip to turn it off when not needed (in sleep mode) – CS_FLASH–LED pin is a CAMERA output pinn that is designed to send ouput signal to a flash to illuminate the surrounding at the exact time of image capture (Will not be used NC) – RESET CIRCUIT – The goal is for the ESP32P4 to reset from global reset pin, but also to reset after a firmware update without causing global reset of other devices. – Implementation: GLOBAL RESET is connected through a buffer to the MCU that only pulls reset LOW and is HIGH–Z otherwise. The MCU RST (PU) pin is exposed with the JTAG pins on a header to allow RST after upload. – PU (RST) Timing Considerations: The PU has to stay low for tSTBL = 50uS after VIN stable at 3.3V – Place a 100nF capacitor at PU.						Peter Hancak Sheet: /Notes/ File: Notes.kicad_sch Title: ESP32P4C5_BOARD Size: A4 Date: 2026–06–01 KiCad E.D.A. 10.0.3		
	1	2	3	4	5	6			